


$$\underline{y} = X\underline{\beta} + \underline{\varepsilon} \quad \underline{\varepsilon} \sim N(0, \sigma^2 I_m)$$

$$\hat{\underline{\beta}}_{MLE} = \hat{\underline{\beta}}_{LS}$$

$$\underline{y} \sim N(X\underline{\beta}, \sigma^2 I_m)$$

$$\hat{\underline{y}} = X\hat{\underline{\beta}} = P_X \underline{y}$$

$$\hat{\underline{\beta}} = (X^T X)^{-1} X^T \underline{y}$$

$$\underline{y} \sim N(X\underline{\beta}, \sigma^2 I_m)$$

$$\underline{z} \sim N(\underline{\mu}, \underline{\Sigma})$$

$$\text{then } A\underline{z} \sim N(A\underline{\mu}, A\underline{\Sigma}A^T)$$

$$\text{so, } \hat{\underline{\beta}} \sim N(\underline{\beta}, \sigma^2 (X^T X)^{-1})$$

$$\underline{e} \sim N(0, (I_m - P_X) \sigma^2)$$

Estimator

$$\text{If } T(\underline{z}) = \sum \alpha_i z_i \rightarrow T(\underline{z}) \text{ linear estimator}$$

$$E(T(z_1, z_2, \dots, z_m)) = 0 \text{ then } T(\underline{z}) \text{ is unbiased estimator}$$

$$\text{ex } E(\hat{\underline{\beta}}) = \underline{\beta} \rightarrow \text{unbiased estimator of } \underline{\beta}$$

$$\hat{\sigma}^2 \text{ is unbiased estimator of } \sigma^2$$

$$\hat{\underline{y}} \text{ \& } \underline{e} \text{ are uncorrelated \& independently distributed when } \underline{\varepsilon} \sim N(0, \sigma^2 I_m) \rightarrow \text{Actual}$$

$$\underline{e} = \underline{y} - \hat{\underline{y}}$$

LUE

$$L^T \underline{y} \text{ is LUE of } P^T \underline{\beta}$$

$$\text{If } E(L^T \underline{y}) = P^T \underline{\beta}$$

LZF

$$\text{If } E(L^T \underline{y}) = 0 \rightarrow \text{LZF}$$

BLUE

$\rightarrow$ LUE with minimum variance

BLUE is uncorrelated with all LZF

$$L^T \underline{y} \rightarrow \text{LUE of } P^T \underline{\beta}$$

$$L^T P_X \underline{y} \rightarrow \text{BLUE of } P^T \underline{\beta}$$

$$SS_{\text{Total}} = \sum (y_i - \bar{y})^2 \rightarrow T \text{ is } \bar{y}$$

$$SS_{\text{error}} = \sum (y_i - \hat{y}_i)^2 \rightarrow \text{Actual} - \text{predicted}$$

$$SS_{\text{model}} = \sum (\hat{y}_i - \bar{y})^2$$

$$SS_{\text{Total}} = SS_{\text{error}} + SS_{\text{model}}$$

Orthogonal polynomial

$$x = (x_1, x_2, \dots, x_m) ; \text{ of } p_0, p_1, \dots, p_k$$

$$p_0(x_i) = 1 \quad \sum p_j(x_i) p_k(x_i) = 0 \quad \forall j \neq k$$

Coefficient of determination

$$R^2 = 1 - \frac{SS_{\text{error}}}{SS_{\text{Total}}}$$

$$R^2 \in (0, 1)$$

Higher value of $R^2 \rightarrow$ implies better model

$R^2 \rightarrow$ increasing function of no. of variables

Adjusted $R^2 \rightarrow$ gives us appropriate time to stop

$$R^2_{\text{adj}} = 1 - \frac{SS_{\text{error}} / (n - k - 1)}{SS_{\text{Total}} / (n - 1)} < R^2$$

$$\rightarrow LTY \rightarrow LUE \quad LTP_x y \rightarrow BLUE \text{ of } p^{TB}$$

BLUE is always unique

For p^{TB} to be estimable $p \in C(X^T)$

$$\rightarrow LTY \text{ is LZF iff } X^T L = 0$$

Time Series

$$\text{cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

→ how much X & Y vary together

If $\text{cov}(X, Y) > 0 \rightarrow X, Y$ increase, decrease together

If $\text{cov}(X, Y) < 0 \rightarrow X \uparrow$ when $Y \downarrow$ & vice versa

Covariance measures joint variability of two random variables

Normalization → scaling method, in which data points are rescaled, so that they end up in a range of 0 to 1

Ex Two different shifts of JEE/CAT exam.

Standardization → scaling method where values are centered around mean with unit standard deviation

$Z = \frac{x - \mu}{\sigma}$ $Z \rightarrow$ standard score $\rightarrow$ represents no. of standard deviations above or below that a specific observation falls

Prediction $\rightarrow$ may or may not be about time

$$y = f(x) \rightarrow \text{given } x \text{ predict } y$$

Forecasting $\rightarrow$ it is about time (There is time involved)

In linear regression $\rightarrow$ answer does not depend on order of data, in time series it does

Time series $\rightarrow$ data given in specific time sequence

"Future looks like the past"

Decomposition $\rightarrow$ decomposes time series in two parts $\rightarrow$ past that is predictable
 $\rightarrow$ " " " not "
 $\hookrightarrow$ ignore

Test situation $\leftarrow y_{t+1} = f(y_t, y_{t-1}, y_{t-2}, \dots)$
Training situation $\leftarrow y_t = f(y_{t-1}, y_{t-2}, y_{t-3}, \dots)$ $\rightarrow$ forecasting the future based on the present is like forecasting the present based on the past

I will build a regression function based on present and past, use it the same regression function to forecast the future, because it is a regression on itself
we call it a Auto regression (Regression on self)

1) Decomposition Method

components of time series

- $\rightarrow$ Trend (long term movements)
- $\rightarrow$ Seasonal component
- $\rightarrow$ Irregular components (Random movements)

white noise $\rightarrow$ Time series from uncorrelated variables

$w_t \sim WN(0, \sigma_w^2) \rightarrow$ need not be normally distributed

Trend $\rightarrow$ A deterministic pattern which persists throughout

$$X_T = T_t + W_t = 0.2t + W_t$$
$$= t^{1/3} + W_t$$

Seasonality $\rightarrow$ Pattern which returns in time series after fixed interval

$$X_T = T_t + S_t + W_t = 0.3t + 3 \sin(4.5t\pi) + W_t$$

Weakly stationary time series

$\rightarrow \mu_t \rightarrow$ constant, independent of t

$\rightarrow \gamma(s, t)$ [ACVF] $\rightarrow$ depends on time lag for $(t, t+h)$

$$\gamma(t+h, t) = \gamma(h, 0)$$

$\hookrightarrow$ independent of t

Strongly stationary time series $\rightarrow$ if distribution remains constant over time

$\{X_t\} \rightarrow$ strongly stationary if shifting the time axis does not affect the distribution

Strongly stationary implies weakly stationary [Not vice versa]

Properties of strongly stationary

- ① X_t are identically distributed
- ② iid sequence is strongly stationary

16/4/23

Random walk with drift (δ)

$$x_t = \delta t + \sum_{j=1}^t w_j \quad t=1, 2, \dots$$

$$\mu_{x_t} = \delta t$$

Auto Covariance function (ACVF) $\rightarrow$ covariance with self

$$\gamma_x(s, t) = \text{cov}(x_s, x_t) = E[(x_s - \mu_s)(x_t - \mu_t)]$$

smooth, so
related

$\uparrow$

Very smooth series $\rightarrow$ ACVF is large, even when t & s are far apart

choppy series $\rightarrow$ ACVF is nearly zero for large separations

ACVF of Random Walk

$$x_t = \sum_{j=1}^t w_j \quad [\text{No drift } \delta=0]$$

$$\gamma_x(s, t) = \min(s, t) \sigma_w^2$$

$\rightarrow$ ACVF of random walk depends on
particular time values of s, t
& not on time separation or lag

$$\text{var}(x_t) = t \sigma_w^2$$

$\hookrightarrow$ increases without bound
as t increases

Autocorrelation function (ACF)

$$\rho(s, t) = \frac{\gamma(s, t)}{\sqrt{\gamma(s, s)\gamma(t, t)}} = \frac{\text{cov}(s, t)}{\text{SD}(s)\text{SD}(t)}$$

$$Z_t \sim \text{iid } N(0, \sigma^2)$$

$$i) X_t = 1 + 2Z_0 \quad (ii) X_t = 1 + 2Z_t + 3Z_{t-2}$$

$$= Z_1 \cos(\theta t) + Z_2 \sin(\theta t)$$

$$i) E(X_t) = 1 + 2 \times 0 = 1$$

$$\gamma(h) = \text{cov}(X_{t+h}, X_t) = E[(X_{t+h} - \mu)(X_t - \mu)]$$

$$= E[(1 + 2Z_0 - 1)(1 + 2Z_0 - 1)]$$

$$= 4Z_0^2 \Rightarrow 16 E(Z_0^2) = 16\sigma^2$$

↳ independent of t

$$(ii) X_t = 1 + 2W_t + 3W_{t-2}$$

$$= 1 + 2W \quad E(X_t) = 1$$

$$E[(2W_{t+h} + 3W_{t+h-2})(2W_t + 3W_{t-2})]$$

$$4W_t W_{t+h} + 6W_t W_{t+h-2} + 6W_{t-2} W_{t+h}$$

$$+ 9W_{t+h-2} W_{t-2}$$

If $h=0$

$$4W_t^2 + 6W_t W_{t-2} + 6W_{t-2} W_t + 9W_{t-2}^2$$

$$16\sigma_w^2 + 81\sigma_w^2 \Rightarrow 97\sigma_w^2$$

If $h \neq 0$ 0

$$iii) X_t = Z_1 \cos(\theta t) + Z_2 \sin(\theta t)$$

$$E(X_t) = E(Z_1 \cos \theta t) + E[Z_2 \sin \theta t]$$

$$= 0$$

$$\gamma(h) = \text{cov}(x_{t+h}, x_t)$$

$$\Rightarrow E \left[z_1 \cos \omega(t+h) + z_2 \sin \omega(t+h) \right. \\ \left. [z_1 \cos \omega t + z_2 \sin \omega t] \right]$$

$$\Rightarrow E \left[z_1^2 \cos \omega t \cos(\omega t + h) + z_1 z_2 \sin \omega t \cos(\omega t + h) \right. \\ \left. + z_2 z_1 \cos \omega(t+h) \sin \omega t + z_2^2 \sin \omega t \times \sin(\omega t + h) \right]$$

$$\Rightarrow E \left[z_1^2 \cos \omega t \cos(\omega t + h) + z_1 z_2 \left[\sin(2\omega t + h) \right. \right. \\ \left. \left. z_1^2 \cos \omega t \cos(\omega t + h) + z_1 z_2 \sin[\omega(2t + h)] \right] \right. \\ \left. + z_2^2 \sin \omega t \sin(\omega t + h) \right]$$

$$\textcircled{2} \quad x_t = \phi x_{t-1} + z_t \quad \underline{|\phi| < 1}$$

$$E(x_t) = E[\phi x_{t-1} + z_t] \\ = \phi E(x_{t-1}) + E(z_t)$$

$$E(x_t) - \phi E(x_{t-1}) \\ = E(z_t)$$

$$\mu = \phi \mu$$

$$\gamma(h) = \text{cov}(x_{t+h}, x_t) = E[x_{t+h} - 0][x_t] \\ = E[x_{t+h} x_t]$$

$$= E[z_{t+h} + \phi x_{t+h-1}][z_t + \phi x_{t-1}]$$

$$\Rightarrow \sum_{h=0}^{\infty} E \left[z_{t+h} \times z_t + \phi z_t x_{t+h-1} + \phi x_{t-1} z_{t+h} + \phi^2 x_{t-1} x_{t+h-1} \right]$$

$$E \left[z_t^2 + \phi z_t x_{t-1} + \phi x_{t-1} z_t + \phi^2 x_{t-1}^2 \right]$$

$$\Rightarrow \sigma_w^2 + \phi^2$$

③ $\{z_t\} \sim \text{i.i.d. } N(0, \sigma^2)$

$$x_t = \begin{cases} z_t & \text{if } t \text{ is even} \\ \frac{z_{t-1}^2 - 1}{\sqrt{2}} & \text{if } t \text{ is odd} \end{cases} \Rightarrow \frac{(z_{t-1} + 1)(z_{t-1} - 1)}{\sqrt{2}}$$

To show $\rightarrow$

$$\{x_t\} \sim WN(0, 1)$$

$$t = 2k$$

$$t = 2k+1$$

$$\sum \left[z_{2k} \right]$$

$$E \left[\frac{z_{2k}^2 - 1}{\sqrt{2}} \right]$$

$$\text{if } t = 2k$$

$$\text{if } t = 2k+1$$

we can handle trends in two ways — detrending
— differencing

↳ does not yield estimate of stationary process

$$x_t - x_{t-1} = (\mu_t - \mu_{t-1}) + (y_t - y_{t-1})$$
$$= \beta_2 + y_t - y_{t-1}$$

$$\nabla x_t = x_t - x_{t-1}$$

First difference $\rightarrow$ eliminates linear trend
second " $\rightarrow$ " quadratic trend

Backshift operator

$$B x_t = x_{t-1}$$

$$B^k x_t = x_{t-k}$$

$$\nabla x_t = (1-B) x_t$$

$$\nabla^2 x_t = (1-B)^2 x_t = (1+B^2-2B) x_t = x_t - 2x_{t-1} + x_{t-2}$$

$$\left[\nabla^2 x_t = \nabla(\nabla x_t) = \nabla(x_t - x_{t-1}) = x_t - x_{t-1} - (x_{t-1} - x_{t-2}) \right. \\ \left. = x_t - 2x_{t-1} + x_{t-2} \right]$$

$$), \quad \nabla^d = (1-B)^d$$

$\rightarrow$ To remove trend with just help of linear regression may not be of much useful, as ACF of residuals of simple linear regression fit to global temp data reveals additional structure in data that regression did not capture

AR(p)

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \dots + \phi_p x_{t-p} + w_t$$

↳ stationary, 0 mean

If mean $\neq 0$

$$x_t - \mu = \phi_1 (x_{t-1} - \mu) + \phi_2 (x_{t-2} - \mu) + \dots + \phi_p (x_{t-p} - \mu) + w_t$$

Here regressors are $\rightarrow$ random components

$$(1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p) x_t = w_t$$

$$\phi(B) x_t = w_t$$

Autoregressive operator is defined as

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$$

AR(1)

$$x_t = \phi x_{t-1} + w_t$$

AR(1) process is same as

$$x_{t-1} = \phi x_{t-2} + w_{t-1}$$

MA(∞) process

$$\vdots$$
$$x_t = \phi^k x_{t-k} + \sum_{j=0}^{k-1} \phi^j w_{t-j}$$

$\rightarrow 0$ as $k \rightarrow \infty$

If $|\phi| < 1$ & x_t is stationary, AR(1) can be represented as a linear process

$$k \rightarrow \infty \quad x_t = \sum_{j=0}^{\infty} \phi^j w_{t-j}$$

$$E(x_t) = E\left(\sum_{j=0}^{\infty} \phi^j w_{t-j}\right) = 0$$

For AR(1) model to be stationary and stable it is necessary that $|\phi| < 1$

$$\gamma(h) = \text{cov}(x_{t+h}, x_t)$$

$$= E\left[\left(\sum_{j=0}^{\infty} \phi^j w_{t+h-j}\right)\left(\sum_{k=0}^{\infty} \phi^k w_{t-k}\right)\right]$$

If $|\phi| > 1 \rightarrow$ series will diverge from its mean [explosive behavior]

$$= E\left[w_{t+h} \dots + \phi^h w_t + \phi^{h+1} w_{t-1} \dots\right] \left[w_t + \phi w_{t-1} + \dots\right]$$

$$= \sigma_w^2 \sum_{j=0}^{\infty} \phi^{h+j} \phi^j = \sigma_w^2 \phi^h \sum_{j=0}^{\infty} \phi^{2j} = \frac{\sigma_w^2 \phi^h}{1 - \phi^2} \quad h \geq 0$$

$$\gamma(h) = \frac{\sigma_w^2 \phi^h}{1 - \phi^2} \quad h \geq 0$$

$$S(h) = \frac{\gamma(h)}{\gamma(0)} = \phi^h \quad h \geq 0$$

↳ satisfies recursion

$$S(h) = \phi S(h-1) \quad h=1, 2, \dots$$

Causal process $\rightarrow$ when a process does not depend on future. ex: AR(1)
when $|\phi| < 1$

MA(q)

$$x_t = w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} - \dots - \theta_q w_{t-q}$$

system is same as infinite moving average defined as linear process

where $\psi_0 = 1$ $\psi_j = \theta_j$ for $j=1, \dots, q$ $\psi_j = 0 \rightarrow$ for other values

$$x_t = \theta(B) w_t$$

$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q$$

MA(1) process

$$x_t = w_t + \theta w_{t-1}$$

$$E(x_t) = 0$$

$$\gamma(h) = \begin{cases} (1+\theta^2)\sigma_w^2 & h=0 \\ \theta\sigma_w^2 & h=1 \\ 0 & h>1 \end{cases}$$

$$S(h) = \begin{cases} \frac{\theta}{1+\theta^2} & h=1 \\ 0 & h>1 \end{cases}$$

In MA(1) $\rightarrow x_t$ is correlated with x_{t-1} , but not with $x_{t-2}, x_{t-3}, \dots$
whereas in AR(1) correlation b/w x_t & x_{t-k} was never zero.

In MA(1) process

$\sigma_w^2 = 1$ & $\theta = 5$ yield same ACF $\gamma(h)$ as $\sigma_w^2 = 25$ & $\theta = 1/5$

so MA(1) process

$$x_t = w_t + \frac{1}{5} w_{t-1} \quad w_t \sim \text{iid } N(0, 25)$$

$$y_t = v_t + 5 v_{t-1} \quad w_t \sim \text{iid } N(0, 5)$$

are same because of normality

ARIMA

order(ARorder, differencing, MAorder)

MA(2) $\Rightarrow$ ARIMA (0,0,2)

AR(1) $\Rightarrow$ ARIMA (1,0,0)

$\rightarrow$ If there's a linear trend $\rightarrow$ 1st difference

" " Quadratic " $\rightarrow$ 2nd difference

Overdifferencing can cause us to introduce unnecessary level of dependency

ARMA(p,q) $\rightarrow$ If it is stationary and

$$x_t = \phi_1 x_{t-1} + \dots + \phi_p x_{t-p} + w_t + \theta_1 w_{t-1} + \dots + \theta_q w_{t-q}$$

p, q $\rightarrow$ orders of autoregressive and moving average

ARMA can be written as

$$\phi(B)x_t = \theta(B)w_t$$

AR and MA polynomials are defined as

$$\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$$

$$\theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$$

$z \Rightarrow$ complex number

To address the problem of future dependent models, we introduce the concept of causality

Causality

ARMA(p,q) $\rightarrow$ is said to be causal, if time series can be written as a one sided linear process i.e.

$$x_t = \sum_{j=0}^{\infty} \psi_j w_{t-j} = \psi(B)w_t$$

$\rightarrow$ Imp

where sum of coefficients i.e. $\sum_{j=0}^{\infty} |\psi_j| < \infty$

ARMA(p,q) $\rightarrow$ is causal if and only if $\phi(z) \neq 0$ for $|z| \leq 1$

$$\psi(z) = \sum_{j=0}^{\infty} \psi_j z^j = \frac{\theta(z)}{\phi(z)} \quad |z| \leq 1 \rightarrow \text{To get coefficients of linear process}$$

other way $\rightarrow$ ARMA(p,q) is causal only when roots of $\phi(z)$ lie outside the unit circle that is $\phi(z)=0$ only when $|z| > 1$

Invertibility

ARMA(p,q) is said to be invertible if it can be written as

$$\pi(B) x_t = \sum_{j=0}^{\infty} \pi_j x_{t-j} = w_t$$

ARMA(b,q) is invertible if and only if $\theta(z) \neq 0$ for $|z| \leq 1$

$$\pi(z) = \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \quad |z| \leq 1$$

ARMA process is invertible only when roots of $\theta(z)$ lie outside unit circle

In causal $\psi(z) = \frac{\theta(z)}{\phi(z)}$ & in invertibility $\pi(z) = \frac{\phi(z)}{\theta(z)}$

$\phi \rightarrow$ for AR
 $\theta \rightarrow$ for MA

For causality $\rightarrow$ check $\phi(z)=0 \rightarrow$ if $|z| > 1$ then causal

For invertibility $\rightarrow$ check $\theta(z)=0 \rightarrow$ if $|z| > 1$ then invertible

To write as linear process, we obtain ψ -weights using

$$\phi(z) \psi(z) = \theta(z)$$

ex $(1-0.9z)(\psi_0 + \psi_1 z + \psi_2 z^2 + \dots) = (1+0.5z)$ $\rightarrow$ ex 3.7 pg 96 for more clarity $\checkmark$

we get $\psi_0 = 1$; $\psi_1 = 0.5 + 0.9 = 1.4$ $\psi_j = 0.9\psi_{j-1}$

$$\text{so } x_t = w_t + 1.4 \sum_{j=1}^{\infty} 0.9^{j-1} \psi_{t-j}$$

PACF

$\hat{x}_{t+h}$ for $h \geq 2$

Regression of x_{t+h} on $\{x_{t+h-1}, x_{t+h-2}, \dots, x_{t+1}\}$

$$\hat{x}_{t+h} = \beta_1 x_{t+h-1} + \beta_2 x_{t+h-2} + \dots + \beta_{h-1} x_{t+1}$$

Similarly

$$\hat{x}_t = \beta_1 x_{t+1} + \beta_2 x_{t+2} + \dots + \beta_{h-1} x_{t+h-1}$$

$$\phi_{hh} = \text{corr}(x_{t+h} - \hat{x}_{t+h}, x_t - \hat{x}_t) \quad h \geq 2$$

Both $x_{t+h} - \hat{x}_{t+h}$ & $x_t - \hat{x}_t$ are uncorrelated with $\{x_{t+1}, \dots, x_{t+h-1}\}$

PACF $\rightarrow \phi_{hh}$ is the correlation b/w x_{t+h} and x_t with linear dependence of $\{x_{t+1}, \dots, x_{t+h-1}\}$ on each removed.

& $\phi_{11} = \phi(1) \rightarrow$ by definition

